

Invoking a System Call

5) the system call function executes and possibly returns a value through a register

prog

stack



usermode

kennebennodde

the 1990s, the number of people in the UK who are employed in the public sector has increased by 1.5 million (1990–1999), and the number of people in the public sector has increased by 2.5 million (1990–1999) (Department of Health 2000).

There is a growing emphasis on the need to improve the efficiency of the public sector, and to ensure that the public sector is able to deliver the best possible value for money. This has led to a number of initiatives, including the introduction of the Health Service Act 1999, which introduced a new framework for the NHS, and the introduction of the Health Service Act 2001, which introduced a new framework for the public sector.

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A speech bubble with a dark gray border and a light gray background. The bubble has a tail pointing towards the bottom left. Inside the bubble, there is a line of code.

...

`write("hello world")`

kernel

process

1) the program calls a library function

lib/write



syscall num

arg #0

2) the library function pushes the syscall number and its arguments onto the stack and triggers a software interrupt

```
write(s):  
    push("write")  
    push(s)  
    int x80
```



x80 interrupt
handler

3) the interrupt handler reads the stack of the interrupted program to extract the system call number and the arguments



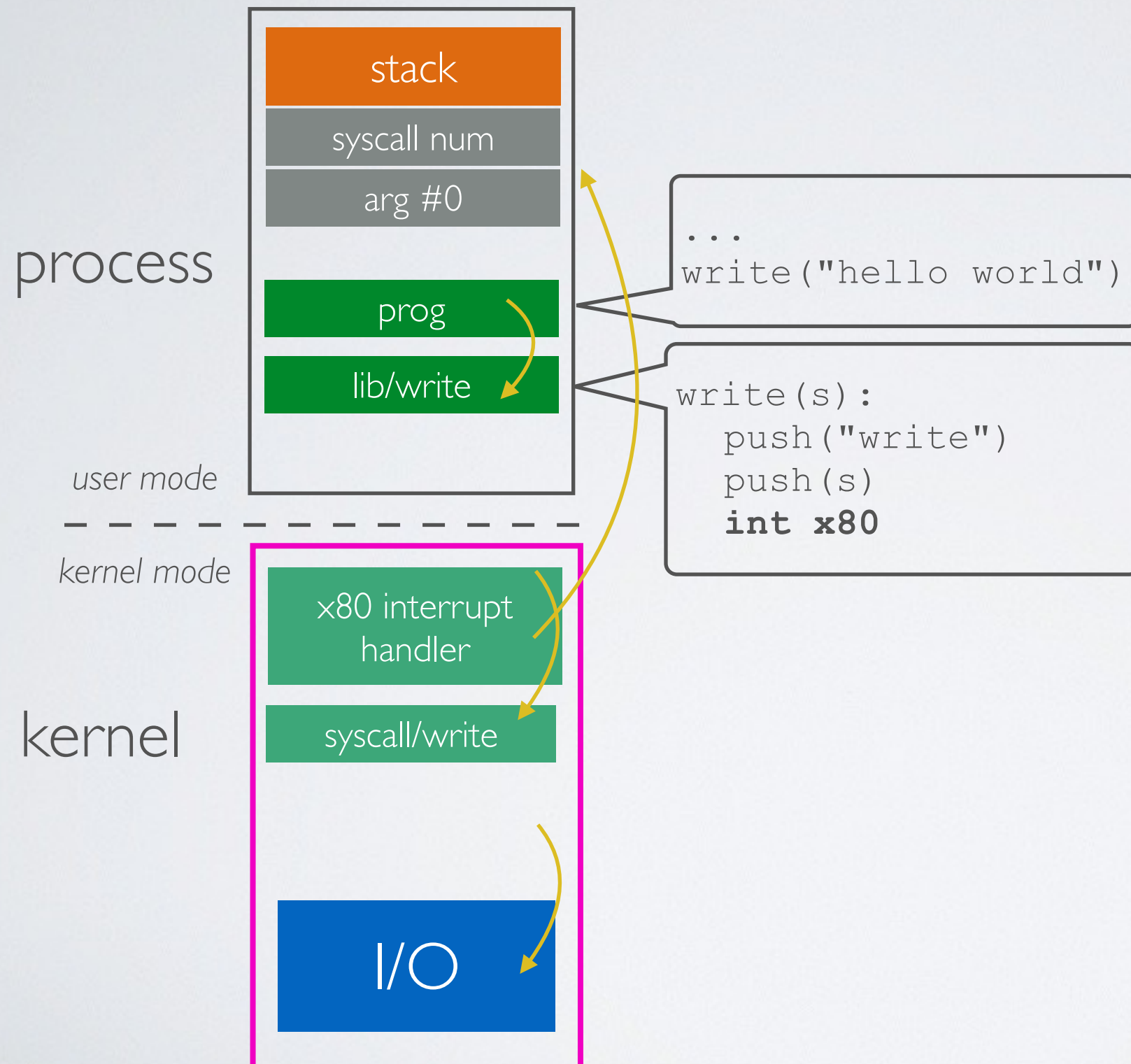
syscall/write

4) the interrupt handler calls the corresponding kernel system call function

1/0



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4) the interrupt handler calls the corresponding kernel system call function

5) the system call function executes and possibly returns a value through a register

Process